quantities we interact with. Once again, the profoundly relational nature of the concepts we use to organize the world emerges.

The cold teaspoon heats up in hot tea because tea and spoon interact with us through a limited number of variables amongst the innumerable variables which characterize their microstate. The value of *these* variables is not sufficient to predict future behaviour exactly (witness the balloon), but is sufficient to predict with optimum probability that the spoon will heat up.

I hope not to have lost the reader's attention with these subtle distinctions …

Now, in the course of the twentieth century thermodynamics (that is, the science of heat) and statistical mechanics (that is, the science of the probability of different motions) have been extended to electromagnetic and quantum phenomena. Extension to include the gravitational field, however, has proved problematic. How the gravitational field behaves when it heats up is still an unsolved problem.

We know what happens to a heated electromagnetic field: in an oven, for instance, there is hot electromagnetic radiation which cooks a pie, and we know how to describe this. The electromagnetic waves vibrate, randomly sharing energy, and we can imagine the whole as being like a gas of photons which move like the molecules in a hot balloon. But what is a hot *gravitational* field?

The gravitational field, as we saw in the first lesson, is space itself, in effect space-time. Therefore when heat is diffused to the gravitational field, time and space themselves must vibrate… But we still don't know how to describe this well. We don't have the equations to describe the thermal vibrations of a hot space-time. What is a vibrating time?

Such issues lead us to the heart of the problem of time: what exactly is the *flow* of time?

The problem was already present in classical physics, and was highlighted in the nineteenth and twentieth centuries by philosophers – but it becomes a great deal more acute in modern physics. Physics describes the world by means of formulae that tell how things vary as a function of 'time'. But we can write formulae that tell us how things vary in relation to their 'position', or how the taste of a risotto varies as a function of the 'variable quantity of butter'. Time seems to 'flow', whereas the quantity of butter or location in space do not 'flow'. Where does the difference come from?